ENGR113 Spring 2025

First-Year Engineering Design

# *Concrete Boat*
# *Final Report*

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**Abstract**

In this project, we designed, built, and tested a small concrete boat to see if material like concrete could float when shaped correctly. Our goal was to create a boat that floats, moves, and can hold some weight. We used computer design tools to make a flat-bottom and wide boat to help it stay balanced on water. We made two types of molds, one by shaping foam with power tools and another using a 3D printer. We picked the best mold based on how easy it was to use and how smooth the boat came out. We mixed the concrete to make it strong and lightweight. We also added a small pull-back motor to a plastic propeller so the boat could move on its own in the water. Some challenges were taking the boat out of the mold without breaking it, making sure the boat did not tip over, and avoiding cracks. The 3D-printed mold made the best boat. It floated well and moved steadily across the water. It was also able to carry some weight, about half its own height. This project helped us learn how to solve problems, work together, and use what we learned in class in a real project. It showed that even heavy materials like concrete can work if designed the right way.

# 1 Introduction

In World War I and World War II, steel shortages caused engineers to look for other resources to create naval ships [2]. What did they turn to? Concrete. 36 concrete ships were made but unfortunately, they were unsuccessful. Some of these ships were sunk to form beachheads. These ships, however, were very strong; sometimes, fleets of ships could not even sink into a concrete ship. The biggest downside of these ships was that they were slow. They could only travel at seven knots, while even aircraft carriers could travel at 25 to 30 knots.

The team is determined to create a prototype concrete boat that is strong, can withstand a heavy load of water intake, and is fast. This prototype will be tested in a tub filled with water, and we will race against other teams to find the best design, mixture of concrete, and dimensions. The goal is to create a functional concrete boat capable of floating and traveling at least a good distance using a simple spring mechanism. This prototype, if successful, could be a blueprint for the Navy to build a better concrete ship that would be roughly two-thirds cheaper than regular steel ships, yet much stronger. Concrete boats were considered the Navy's worst idea, but with this research, it could be a game changer. [3]

## *1.1 Problem Overview*

Creating a floating concrete boat involves several challenges and problems. Since concrete is very dense and heavy, we need to create a light boat design and be able to hold a significant amount of water. One requirement of the boat is that the boat may not go over the dimensions of 8" x 6" x 3". It also has to be lightweight and have a wide hull for up thrust. The use of thin walls is also of great importance in reducing weight. For the boat to have stability, it must have a symmetrical shape. For the demolding process to go smoothly, the molds for the concrete boat must be smooth to prevent any irregular areas. Lastly, we want to make sure it travels quickly and in a straight line.

### *1.2 Existing Solutions*

To ensure our concrete boat could float effectively, we opted for a wide, flat-bottomed design with thin walls, maximizing surface area and allowing greater upward buoyant force. To facilitate clean demolding without damaging the structure, we applied tape and sprayed WD-40 on the mold surface, which provided a smooth release interface. Our mold materials—foam and 3D-printed plastic—were selected for ease of shaping, consistency, and accuracy. The entire design process was carried out in SolidWorks to ensure precise geometry and uniform wall thickness. We thoroughly sanded the mold to achieve a smooth surface finish on the final boat, reducing drag and potential weak points. To enhance durability, we extended the concrete curing time, preventing cracks and improving structural integrity.

Post-curing, the boat was lightly sanded to trim excess weight and improve buoyancy. For propulsion, we tested multiple prototypes and implemented a pull-back spring motor with a plastic propeller. This allowed the boat to move swiftly and in a straight line, demonstrating both stability and control. All design choices prioritized a balance between strength, weight, and hydrodynamic performance, enabling the boat to be light enough for spring-powered movement while maintaining structural reliability in water.

### *1.3 Project Objectives*

Our project was to design and build a small concrete boat capable of floating and moving 6 to 10 feet across water. We developed a lightweight yet strong concrete mix and built a simple wind-up spring propeller system for propulsion. The goal was to make a typically heavy material, like concrete, work in a floating structure.

The boat was engineered to avoid cracking, tipping, or sinking, and was expected to move at around 1 inch per second while carrying at least half its own weight. In the end, we presented a fully functional concrete boat along with a detailed report outlining our design process, technical work, and key takeaways. The project applied core engineering principles while strengthening our teamwork and problem-solving skills.

# 2 Technical Activities

## *2.1 Mechanical Test*

Our formula for the concrete paste was 300 grams of Quikrete with 2 grams of polystyrene foam bean, creating a W/C ratio of 0.34. This paste was poured into four cylinders and were used in a series of compression and split tensile tests. They were tested physically with a machine and mathematically with equations.

We created two stress-strain graphs for each test. The stress-strain graph gave us the point where the concrete withstands its peak stress, the point just before it breaks and cracks. The stress and strain value of the points are recorded below in the table to complete our mechanical test.

**Table 1: Mechanical Test Results**

| Physical Property | Compressive Test | Tensile Test |
|---|---|---|
| Maximum Stress (PSI) | 767.21 | 280.1 |
| Maximum Strain (%) | 2.6 | 1.14 |

## *2.2 Project Budget*

A complete budget helped create the two molds. Also, obtaining the pull-back mechanism to create a mechanically propelled boat was also bought. Below is the list of materials purchased for mold construction and the propeller system.

**Table 2: Project budget**

| **Item** | **Cost ($)** |
|---|---|
| Expanding Spray Foam | $4.00 |
| 3D Printer Filament PLA | $8.00 |
| Propeller pullback mechanism | $5.00 |
| **Total Cost** | **$17.00** |

### *2.3 Project Timeline*

Overview of the concrete boat project covering design, fabrication, and testing phases, with a week-by-week breakdown of completed and planned tasks.

**Table 3: Project Timeline**

| Task | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Draft Boat Design | x | x | x | | | | | | | |
| Cement samples testing | | x | x | x | | | | | | |
| Designing the 3D mold | | | x | x | x | | | | | |
| Designing the propeller | | | x | x | x | x | | | | |
| Creating the foam mold | | | | x | x | x | | | | |
| Printing the 3D mold | | | | | | x | | | | |
| Demolding both molds and sanding | | | | | | | x | x | | |
| Making the propeller | | | | | | | x | x | x | |
| Testing of the Concrete boat | | | | | | | x | x | x | |
| Final Report | | | | | | | x | x | x | x |
| Summarizing Presentation | | | | | | | | x | x | x |

## 3 Results

The testing phase was used to evaluate different properties of the concrete mixture, the workability with the two types of molds, and the overall performance of the concrete boats, including the propeller system. We researched different approaches and went ahead with the one with the easiest workability and performance boost. [5]

### *3.1 Properties of Cementitious Material*

The primary construction material used for the boat was cement. A concrete paste was prepared by combining cement and water at a water-to-cement (W/C) ratio of 0.30. This ratio was determined to be optimal after conducting multiple trial mixes. The resulting paste exhibited favorable workability characteristics, maintaining a balanced viscosity that was both easy to handle and suitable for pouring into the mold. This consistency was critical in achieving a uniform fill and reducing the likelihood of air pockets or segregation during casting. [4]

### *3.2 Mold for the cement boat*

| | |
|---|---|
| | |
| Figure 1: Primary mold (3d printed mold) | Figure 2: Secondary mold (Foam mold) |

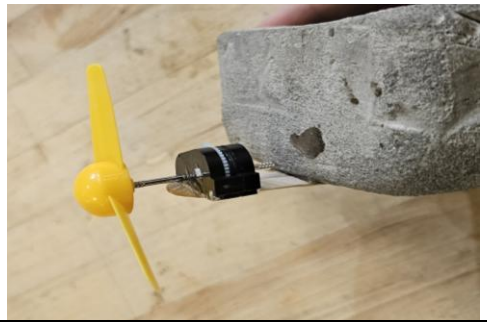
After demolding both the secondary (Figure 2) and primary (Figure 1) molds, the 3D printed mold proved to be the superior option, even though the foam mold was easier to shape [1]. It precisely modeled surface curvature and consistent offset between the inner and outer shells resulted in a uniform wall thickness and symmetrical final shape. Additionally, the concrete paste poured smoothly into the mold and settled evenly, contributing to a clean, well-defined structure with minimal surface defects.

### *3.3 The cement boat*

| | |
|---|---|
| | |
| Figure 3: Primary boat created from a 3D printed mold | Figure : Secondary boat created from XPS insulation foam mold |

The final boats we got from the molds were sanded down and repaired to the desired shape, structure, and weight. This enabled us to achieve optimal performance and maintain buoyancy in the water.

### *3.4 The Propeller System*

| | |
|---|---|
| |  |
| Figure 7: Pull-back mechanism | Figure 8: Propeller mount |

The team explored multiple propulsion concepts before concluding that a simple, lightweight design would minimize additional weight and preserve the boat's stability. The final mechanism consists of a gear-driven propeller shaft powered by a wind-up spring. This stored mechanical energy is transferred through the gear system. The propeller, mounted at the end of the shaft, spins in a clockwise direction to generate forward thrust and propel the boat efficiently.

### *3.5 Final Performance*

The 3D-printed mold boat was selected as the final design due to its aerodynamic shape, lightweight structure, symmetrical curves, and smooth edges. It reached a speed of 10.4 inches per second and moved in a straight line. The boat weighed 482g and successfully held 210g of water, resulting in a water-to-boat weight ratio of 0.44. This demonstrated that the boat could carry nearly half its weight before fully submerging.

## 4 Discussion

We faced some challenges during the course of our project. In molds, shaping the foam mold by hand was hard and did not always give us the required results. We added spray foam to fill in gaps and add to the foam to sand down to perfect shape. The 3D-printed mold gave a much better shape but needed a lot of time to design and print. To avoid the concrete getting stuck to the XPS foam, we had to use tape and WD-40. Finding the right concrete mix was also difficult; the first mixes we used for the foam mold and the mechanical tests were too thick and hard to pour, but the mix for the 3D-printed mold worked much better. Taking the boat out of the mold without

cracks was a challenge and needed careful handling. Teamwork was important when splitting tasks, and we had to communicate clearly to keep things on track. Time management helped us stay organized and finish all parts of the project on time. Through this project, we learned how to solve real problems, work as a team, and apply what we learned in engineering class to hands-on work.

## 5 Conclusion

The Concrete Boat project was a valuable learning experience that allowed our team to apply core engineering skills in a hands-on, collaborative setting. From the initial design phase to final testing, we engaged in every step of the engineering process, including material selection, mold creation, structural testing, and propulsion design. One of the biggest challenges we faced was balancing the density and strength of concrete with the need for buoyancy. Through trial and error, we experimented with different concrete mixes and mold materials, ultimately discovering that the 3D-printed mold provided the most accurate and consistent results. It allowed us to achieve a smooth, symmetrical boat with thin walls and minimal surface defects, improving both appearance and performance. We also designed a simple yet effective pull-back propulsion system that helped move the boat across the water without adding extra weight.

In addition to the technical skills we developed, this project emphasized the importance of teamwork, communication, and time management. Splitting tasks efficiently and solving problems together helped us stay on schedule and complete the project successfully. It also showed us how to connect classroom concepts to real-world engineering challenges and adapt when things didn’t go as planned. By the end of the quarter, we had not only built a functional concrete boat but also gained a deeper understanding of the design process and the value of persistence, creativity, and collaboration in engineering.

References

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